

Generative AI - 50 Interview Questions with Answers

Generative AI Fundamentals (Questions 1-10)

Q1. What is Generative AI and how does it differ from traditional AI?

Answer:

Generative AI: Creates new content (text, images, code, audio).

Traditional AI: Classifies, predicts, or analyzes existing data.

Comparison:

Aspect	Traditional AI	Generative AI
Output	Classification, prediction	New content creation
Examples	Spam detection, fraud detection	ChatGPT, DALL-E, Midjourney
Training	Supervised learning	Self-supervised, unsupervised
Goal	Pattern recognition	Pattern synthesis

Examples:

- **Traditional:** "Is this email spam?" (Yes/No)
- **Generative:** "Write an email about..." (Creates email)

Q2. Explain the difference between discriminative and generative models.

Answer:

Discriminative models: Learn $P(Y|X)$ - probability of Y given X

Input X → Model → Class Y

Example: Given image, classify as cat or dog

Generative models: Learn $P(X|Y)$ or $P(X)$ - distribution of data

Noise/Prompt → Model → Generate X

Example: Generate image of a cat

Examples:

- **Discriminative:** Logistic Regression, SVM, CNNs for classification
- **Generative:** GANs, VAEs, GPT, Diffusion models

Mathematical:

- Discriminative: Directly model decision boundary
- Generative: Model data distribution, can sample new examples

When to use:

- **Discriminative:** Classification tasks, often better performance
- **Generative:** Need to create new data, understand data distribution

Q3. What are the main types of generative models?

Answer:**1. Autoregressive Models:**

- Generate sequentially, one element at a time
- Examples: GPT, PixelCNN

$$P(x) = P(x_1) \times P(x_2 | x_1) \times P(x_3 | x_1, x_2) \times \dots$$

2. Variational Autoencoders (VAEs):

- Encode to latent space, decode to generate
- Continuous latent space
- Good for interpolation

3. Generative Adversarial Networks (GANs):

- Generator vs Discriminator
- High-quality images
- Training instability

4. Diffusion Models:

- Add noise, then denoise
- State-of-the-art image generation
- Examples: DALL-E 2, Stable Diffusion, Midjourney

5. Transformer-based:

- Attention mechanism
- Examples: GPT, BERT, T5

6. Flow-based models:

- Invertible transformations
- Exact likelihood computation

Q4. Explain how GPT (Generative Pre-trained Transformer) works.

Answer:**Architecture:** Decoder-only Transformer**Training:**

1. **Pre-training:** Predict next token

```
Input: "The cat sat on the"  
Target: "mat"
```

2. Trained on massive text corpus (internet, books)
3. Self-supervised learning

Key components:**1. Tokenization:**

```
"Hello world" → [15496, 995]
```

2. Embeddings:

- Token embeddings
- Positional embeddings

3. Transformer blocks:

- Multi-head self-attention
- Feed-forward networks
- Layer normalization

4. Generation:

```
Prompt → Token probabilities → Sample → Repeat
```

Versions:

- GPT-1: 117M parameters
- GPT-2: 1.5B parameters
- GPT-3: 175B parameters
- GPT-4: Estimated 1.7T+ parameters (rumored)

Capabilities:

- Text generation
- Translation
- Summarization

- Question answering
- Code generation
- Few-shot learning

Q5. What are Generative Adversarial Networks (GANs)?

Answer:

Architecture: Two neural networks competing

Components:

1. Generator (G):

- Creates fake data
- Input: Random noise (latent vector)
- Output: Synthetic data

$z \text{ (noise)} \rightarrow \text{Generator} \rightarrow \text{Fake image}$

2. Discriminator (D):

- Distinguishes real from fake
- Input: Real or fake data
- Output: Probability (real or fake)

$\text{Image} \rightarrow \text{Discriminator} \rightarrow [0, 1]$

Training (Minimax game):

$$\min_G \max_D V(D, G) = E[\log D(x)] + E[\log(1 - D(G(z)))]$$

Process:

1. Generator creates fake data
2. Discriminator evaluates real vs fake
3. Both networks improve iteratively
4. Generator gets better at fooling discriminator
5. Discriminator gets better at detecting fakes

Types:

- **DCGAN:** Deep Convolutional GAN
- **StyleGAN:** Style-based generator
- **CycleGAN:** Image-to-image translation
- **Pix2Pix:** Paired image translation

Applications:

- Image generation
- Style transfer
- Super-resolution
- Data augmentation

Challenges:

- Training instability
- Mode collapse
- Vanishing gradients

Q6. Explain diffusion models.

Answer:

Concept: Add noise gradually, then learn to denoise

Process:**Forward (Diffusion):**
$$x_0 \text{ (clean)} \rightarrow x_1 \rightarrow x_2 \rightarrow \dots \rightarrow x_t \text{ (pure noise)}$$

Add Gaussian noise at each step

Reverse (Denoising):
$$x_t \text{ (noise)} \rightarrow \dots \rightarrow x_1 \rightarrow x_0 \text{ (clean image)}$$

Neural network learns to remove noise

Training:

1. Take real image
2. Add random amount of noise
3. Train model to predict noise
4. Loss: MSE between predicted and actual noise

Generation:

1. Start with pure noise
2. Iteratively denoise
3. 1000+ steps typically
4. Guided by text prompt (CLIP)

Types:

- **DDPM:** Denoising Diffusion Probabilistic Models
- **DDIM:** Faster sampling
- **Latent Diffusion:** Work in latent space (Stable Diffusion)

Models:

- DALL-E 2
- Stable Diffusion
- Midjourney
- Imagen

Advantages:

- High-quality images
- Stable training
- Diverse outputs

Disadvantages:

- Slow generation (many steps)
- Computationally expensive

Q7. What is a Variational Autoencoder (VAE)?

Answer:

Architecture: Encoder → Latent space → Decoder

Components:

1. Encoder:

- Maps input to latent distribution
- Outputs: μ (mean), σ (std deviation)

$x \rightarrow \text{Encoder} \rightarrow [\mu, \sigma]$

2. Latent space:

- Sample $z \sim N(\mu, \sigma)$
- Reparameterization trick: $z = \mu + \sigma \times \epsilon$

3. Decoder:

- Reconstructs input from latent

$z \rightarrow \text{Decoder} \rightarrow x'$

Loss function:

$$\begin{aligned}\text{Loss} &= \text{Reconstruction loss} + \text{KL divergence} \\ &= ||x - x'||^2 + \text{KL}(q(z|x) || p(z))\end{aligned}$$

Training:

- Minimize reconstruction error
- Keep latent distribution close to $N(0,1)$

Generation:

1. Sample z from $N(0,1)$
2. Pass through decoder
3. Get new data point

Properties:

- **Continuous latent space:** Can interpolate
- **Smooth transitions:** Between data points
- **Interpretable dimensions:** Sometimes

Applications:

- Image generation
- Data compression
- Anomaly detection
- Drug discovery

vs GAN:

- VAE: Blurrier but stable training
- GAN: Sharper but unstable

Q8. What is prompt engineering?

Answer:

Definition: Crafting inputs to get desired outputs from LLMs

Techniques:**1. Zero-shot:**

```
"Translate to French: Hello world"
```

2. Few-shot:

```
"dog → animal  
car → vehicle"
```

```
apple → ?"
```

3. Chain-of-thought:

```
"Let's solve this step by step:  
1. First...  
2. Then...  
3. Finally..."
```

4. Role prompting:

```
"You are an expert Python developer.  
Write a function to..."
```

5. Instruction clarity:

```
Bad: "Write about dogs"  
Good: "Write a 200-word article about dog training for puppies, including 3  
specific techniques"
```

6. Constraints:

```
"Answer in exactly 3 sentences.  
Use simple language suitable for a 10-year-old."
```

7. Format specification:

```
"Respond in JSON format:  
{  
  "answer": "...",  
  "confidence": 0.0-1.0  
}"
```

Best practices:

- Be specific and clear
- Provide context
- Use examples
- Iterate and refine
- Test variations

Q9. Explain text-to-image generation.

Answer:

How it works:

1. Text encoding (CLIP):

- Text → Text encoder → Text embedding

2. Image generation:

- Text embedding → Image generator → Image
- Typically diffusion model

3. Iterative refinement:

- Start with noise
- Guided by text embedding
- Denoise step by step

Architecture (Stable Diffusion):

```
Text → CLIP encoder → Conditioning →  
Noise → U-Net (denoise) → Latent image →  
VAE decoder → Final image
```

Key components:

CLIP (Contrastive Language-Image Pre-training):

- Aligns text and images
- Trained on 400M text-image pairs

Latent Diffusion:

- Work in compressed space
- Faster than pixel-space diffusion

Classifier-free guidance:

- Control image quality vs diversity
- Higher guidance = more faithful to prompt

Prompting techniques:

```
"A photorealistic portrait of a cat,  
highly detailed, 8k, trending on artstation,  
digital art, dramatic lighting"
```

Models:

- DALL-E 2/3: OpenAI
- Stable Diffusion: Stability AI
- Midjourney: Independent
- Imagen: Google

Q10. What is Retrieval Augmented Generation (RAG)?**Answer:**

Problem: LLMs have knowledge cutoff, can hallucinate

Solution: Retrieve relevant info, then generate

Architecture:

Query → Retrieve docs → Augment prompt → LLM → Response

Components:**1. Document processing:**

- Chunk documents (512-1024 tokens)
- Generate embeddings
- Store in vector database

2. Retrieval:

- Query → Embedding
- Semantic search
- Top-k most relevant chunks

3. Augmentation:

```
Prompt: "Answer based on context:  
Context: {retrieved_docs}  
Question: {question}  
Answer:"
```

4. Generation:

- LLM generates with context

Benefits:

- Up-to-date information
- Reduced hallucinations
- Factual accuracy

- Transparency (cite sources)

Challenges:

- Retrieval quality
- Context length limits
- Latency
- Cost

Tools:

- LangChain
- LlamaIndex
- Vector DBs: Pinecone, Weaviate, Chroma

Image Generation (Questions 11-20)

Q11. Explain DALL-E and how it works.

Answer:**DALL-E 1:**

- Modified GPT-3
- Treats images as sequences of tokens
- Autoregressive generation

DALL-E 2:

- **CLIP**: Text-image alignment
- **Prior**: Text → Image embedding
- **Decoder**: Diffusion model
- High resolution (1024×1024)

DALL-E 3:

- Better prompt following
- Integrated with ChatGPT
- Improved quality and detail

Process:

1. Text prompt
2. CLIP encodes text
3. Prior predicts image embedding
4. Diffusion decoder generates image
5. Super-resolution upscaling

Capabilities:

- Object combinations
- Styles (photorealistic, painting, etc.)

- Variations on existing images
- Inpainting/outpainting

Q12. What is Stable Diffusion?

Answer:

Key innovation: Latent diffusion model

Architecture:

1. VAE encoder:

- Image → Compressed latent (8x smaller)

2. U-Net:

- Denoise in latent space
- Conditioned on text (CLIP)

3. VAE decoder:

- Latent → Image

Advantages:

- **Open-source:** Free to use
- **Efficient:** Runs on consumer GPUs
- **Fast:** ~50 steps vs 1000
- **Controllable:** Many parameters

Versions:

- SD 1.5: 512×512
- SD 2.0/2.1: 768×768
- SDXL: 1024×1024

Techniques:

ControlNet: Add spatial guidance

Pose, depth, canny edges → Control generation

LoRA: Fine-tune on specific styles

DreamBooth: Personalization

Textual Inversion: Learn new concepts

Q13-20. [More image generation topics]

Q13: Midjourney and its features **Q14:** Image inpainting and outpainting **Q15:** Style transfer with GANs **Q16:** Super-resolution models **Q17:** Image-to-image translation **Q18:** ControlNet and guidance **Q19:** Negative prompts **Q20:** Prompt engineering for images

Text Generation (Questions 21-30)

Q21. Explain ChatGPT and its training process.

Answer:

Training stages:

1. Pre-training (GPT base model):

- Next token prediction
- Internet text corpus
- Unsupervised learning

2. Supervised Fine-Tuning (SFT):

- Human-written conversations
- Instruction-following examples
- ~10K high-quality conversations

3. Reward Model (RM):

- Human rankers compare outputs
- Train model to predict preferences
- Scalar reward for each response

4. Reinforcement Learning (PPO):

- Optimize for reward
- Balance: High reward + stay close to SFT
- Proximal Policy Optimization

Result: Helpful, harmless, honest assistant

Versions:

- GPT-3.5-turbo: Fast, affordable
- GPT-4: More capable, reasoning
- GPT-4-turbo: Longer context (128K)

Q22. What is instruction tuning?

Answer:

Process: Fine-tune on diverse instruction-following tasks

Dataset format:

```
{
  "instruction": "Translate to Spanish",
  "input": "Hello world",
  "output": "Hola mundo"
}
```

Datasets:

- **FLAN:** 1,800+ tasks
- **InstructGPT:** Human feedback
- **Alpaca:** 52K instructions from GPT-3.5
- **Dolly:** 15K human-generated

Benefits:

- Better zero-shot performance
- Follows instructions better
- Generalizes to new tasks
- More helpful outputs

Models:

- InstructGPT
- FLAN-T5
- Alpaca
- Vicuna

Q23-30. [More text generation]

Q23: Few-shot learning in LLMs **Q24:** Temperature and sampling strategies **Q25:** Beam search vs sampling
Q26: Top-k and top-p (nucleus) sampling **Q27:** Model fine-tuning vs prompting **Q28:** LoRA for efficient fine-tuning **Q29:** Model compression and quantization **Q30:** Multi-turn conversation management

Code Generation (Questions 31-35)

Q31. Explain GitHub Copilot and code generation models.

Answer:**GitHub Copilot:**

- Powered by OpenAI Codex
- Trained on public code repositories
- Suggests code as you type

Codex:

- Based on GPT-3
- Fine-tuned on code (GitHub, Stack Overflow)
- Understands multiple languages

Capabilities:

- Autocomplete
- Generate functions from comments
- Explain code
- Fix bugs
- Write tests

Other code models:

- **Code Llama**: Meta's code LLM
- **StarCoder**: Open-source code model
- **AlphaCode**: DeepMind's competitive programming
- **Replit Ghostwriter**: AI pair programmer

Q32-35. [More code generation]

Q32: Code LLaMA architecture **Q33**: Prompt engineering for code **Q34**: Code evaluation and HumanEval **Q35**: AI-assisted debugging

Applications & Ethics (Questions 36-45)

Q36. What are the main applications of Generative AI?

Answer:**Content Creation:**

- Articles, blog posts
- Marketing copy
- Social media content
- Email drafts

Creative Arts:

- Digital art, illustrations
- Music composition
- Video generation
- Game assets

Software Development:

- Code generation
- Code review
- Documentation
- Bug fixing

Business:

- Customer service chatbots
- Report generation

- Data analysis
- Presentation creation

Education:

- Tutoring
- Quiz generation
- Personalized learning
- Homework help

Research:

- Literature review
- Hypothesis generation
- Drug discovery
- Protein folding

Healthcare:

- Medical imaging
- Drug design
- Diagnosis assistance
- Treatment planning

Q37. What are the ethical concerns with Generative AI?

Answer:**1. Misinformation:**

- Fake news generation
- Deepfakes
- Propaganda

2. Bias and Fairness:

- Training data biases
- Demographic disparities
- Stereotypes

3. Copyright and IP:

- Training on copyrighted material
- Who owns generated content?
- Artist attribution

4. Job Displacement:

- Writers, artists, programmers
- Economic impact
- Skill devaluation

5. Privacy:

- Model memorization
- PII leakage
- Consent for training data

6. Misuse:

- Phishing emails
- Academic dishonesty
- Scams and fraud

7. Environmental:

- Energy consumption
- Carbon footprint
- Resource usage

Mitigation:

- Watermarking
- Detection tools
- Ethical guidelines
- Regulation
- Transparency

Q38-45. [More applications and ethics]

Q38: Deepfakes and detection **Q39:** Content moderation challenges **Q40:** Watermarking generated content
Q41: AI detection tools **Q42:** Copyright issues **Q43:** Bias in generative models **Q44:** Environmental impact
Q45: Regulation and governance

Technical Deep Dives (Questions 46-50)

Q46. How do you evaluate generative models?

Answer:

Image generation:**1. Quantitative:**

- **FID** (Fréchet Inception Distance): Lower is better
- **IS** (Inception Score): Higher is better
- **LPIPS**: Perceptual similarity
- **SSIM**: Structural similarity

2. Qualitative:

- Human evaluation
- Visual inspection
- User studies

Text generation:**1. Automatic metrics:**

- **Perplexity:** Lower is better
- **BLEU/ROUGE:** For specific tasks
- **BERTScore:** Semantic similarity

2. Human evaluation:

- Coherence
- Relevance
- Fluency
- Factual accuracy

Code generation:

- **pass@k:** Percentage passing tests
- **HumanEval:** Benchmark dataset
- **MBPP:** Python programming problems

General:

- Diversity
- Novelty
- Faithfulness to prompt
- Safety

Q47. What is the attention mechanism in Transformers?

Answer:

Purpose: Focus on relevant parts of input

Self-attention:

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) \times V$$

Process:

1. Create Query, Key, Value matrices
2. Compute attention scores ($Q \times K^T$)
3. Scale by $\sqrt{d_k}$
4. Apply softmax
5. Multiply by Values

Multi-head attention:

- Multiple attention mechanisms in parallel
- Each "head" learns different patterns

- Concatenate and project

Example:

Input: "The cat sat on the mat"

Self-attention learns:

- "cat" attends to "sat" (subject-verb)
- "sat" attends to "mat" (verb-object)
- "on" attends to "mat" (preposition-object)

Types:

- **Encoder:** Bidirectional (BERT)
- **Decoder:** Causal (GPT)
- **Cross-attention:** Encoder-Decoder

Q48-50. [Final technical topics]

Q48: Training large models (distributed, mixed precision) **Q49:** Model compression and distillation **Q50:** Future directions in Generative AI

Quick Interview Tips

1. **Understand fundamentals:** GANs, VAEs, Transformers, Diffusion
 2. **Know popular models:** GPT, DALL-E, Stable Diffusion, Midjourney
 3. **Practical experience:** Mention tools used (OpenAI API, HuggingFace, Stable Diffusion)
 4. **Ethics awareness:** Discuss bias, misuse, copyright
 5. **Applications:** Real-world use cases
 6. **Technical depth:** Attention, training processes
 7. **Current trends:** LLMs, multimodal models, efficient training
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Good luck with your Generative AI interview! 🚀